Chapter 5

Working with AI in a Roundup of Business Disciplines

IN THIS CHAPTER

» **Comparing old with new processes**

» **Guarding against hallucinations and errors**

» **Grasping the many uses of GenAI across industries**

This chapter introduces you to some of the changes that will ripple through your work life no matter what industry or profession you're in. Of course, I can't cover every profession and industry in this one book, but this chapter covers enough to give you an idea of the versatility, flexibility, and usefulness of GenAI in various disciplines.

Using GenAI for Marketing

GenAI can be a powerful tool in marketing, where creative thinking and tasks mesh with data-driven campaigns. Unfortunately, many users employ GenAI as a shortcut in dealing with content churn, meaning the rate or frequency at which published content is replaced, removed, or significantly updated over time, rather than a way to improve the performance of marketing campaigns and efforts. That's understandable to a degree, given that any job in marketing is typically an underappreciated and overworked position. On any given day, a client or stakeholder confuses marketing with advertising and mistakes metrics for actual human connections.

You use GenAI to fill content buckets, a term that represents a topic or pillar around which content is created and published, and content funnels, which refers to the route marketing takes a potential customer (or reader) from initial awareness of your brand to ultimately converting to a purchase. But if that's all you use GenAI for, the results tend to be less than stellar and (worse) can add to the perception that marketing jobs can be fully automated. This perception, of course, is not the case in reality.

Creating general to detailed content

One area in which GenAI can be particularly useful is in content creation. It can generate marketing content ranging from general information to detailed sales content. GenAI can be helpful with the entire spectrum of content, from research reports and white papers, product information, and fact sheets to blog posts, social media updates, and email newsletters.

Another valuable application of GenAI is for customer engagement. You can integrate it into customer service platforms to handle common inquiries, freeing up human agents to tackle more complex issues. For instance, an e-commerce website can use GenAI to answer frequently asked questions about shipping policies, return procedures, and product details, providing instant responses and improving customer satisfaction.

If you prefer not to use GenAI as a customer-facing chatbot, you can use it to generate responses that a traditional, rule-based chatbot (such as Alexa or Siri) can then deliver to customers. This approach leverages the strength of GenAI (content or information generation) with the strength of traditional chatbots (reliably and correctly serving customers the correct "canned response"). This method eliminates common problems with GenAI, such as AI hallucinations and accidental data leaks.

Analyzing customer data

GenAI can also assist in market research by analyzing customer feedback and reviews. By processing large volumes of unstructured text data, it can identify common themes and sentiments, helping marketers understand customer preferences and pain points. For example, a tech company can

use GenAI to sift through product reviews and social media comments to gather insights on user experience, which can then inform product development and marketing strategies. Typically, you need to add that data to retrieval-augmented generation (RAG), attach a text, CSV, or Excel file to a prompt, or add the data in your prompt if it isn't too large.

GENAI IN MARKETING

Pros	Cons
Efficiency in content creation	May require oversight to ensure accuracy
Quickly generates content	Can produce generic or repetitive outputs
Enhanced customer engagement	Limited understanding of nuanced queries
Provides instant responses to common inquiries	May struggle with complex customer issues
Personalization	Data privacy concerns
Tailors messages based on customer data	Requires careful handling of customer data
Market research	Interpretation challenges
Analyzes large volumes of text data for insights	May misinterpret subtle sentiments
Cost-effective	Initial setup and integration effort

Reduces the need for extensive human resources	Requires time and resources to implement
Creativity boost	Dependence risk
Offers diverse ideas and perspectives	Over-reliance may stifle human creativity

For personalized marketing, you can use GenAI to tailor messages based on customer data. By leveraging customer profiles and purchase history, GenAI can generate personalized recommendations and offers. For instance, an online bookstore can use GenAI to send personalized emails suggesting new books based on a customer's previous purchases and browsing history. Typically, this requires integrating GenAI with your customer relationship management (CRM), recommendation engine, or similar system.

Retrieving Smart Answers for HR

Human Resources (HR) can use GenAI to good effect in many ways. If you're in HR, you can use it to create content for a self-help repository to handle routine inquiries from employees, such as questions about company policies, benefits, and procedures. By doing so, you can free up time to focus on more complex and strategic tasks. For example, if an employee wants to know how to apply for parental leave, GenAI can provide the necessary steps and direct them to the appropriate forms or contacts.

Other applications for GenAI in the HR department include

- **Assisting in the recruitment process by handling initial candidate interactions.** When a candidate applies for a job, GenAI can engage with them to gather basic information, answer frequently asked questions about the role, and even schedule interviews. This not only speeds up the process but also ensures that candidates receive timely and consistent communication, which in turn creates a good impression of your company.
- **Cataloging training and development opportunities by generating scripts and employing AI video generator apps.** GenAI can point to or provide em-

ployees with resources and answer questions related to training programs. For example, if an employee wants to enhance their project management skills, GenAI can suggest relevant training programs, provide a list of training videos, suggest credentials they might want to pursue, and offer detailed information on course availability and steps to enroll.



REMEMBER Involving AI in HR activities isn't a set-and-forget exercise; however, HR departments must regularly update a GenAI tool with current company information and policies. You can do this by

- **Asking IT or your company's AI department to retrain or fine-tune the model,** or add updated and remove old information in Retrieval Augmented Generation (RAG). See Book 2, Chapter 6 to learn about RAG. Otherwise, response accuracy and relevance can drift off course over time.
- **Monitoring the interactions between GenAI and employees to ensure GenAI is performing well.** You can also identify any need for further training or adjustments to the model if it isn't performing to your expectations. Typically, HR departments do this by periodically reviewing chat logs to spot response errors and gathering feedback from employees about their experiences with GenAI.

COMPARATIVE TABLE OF PROS AND CONS OF USING GENAI IN HR TASKS

Pros	Cons
Efficiency: Automates routine inquiries, freeing up HR staff for more complex tasks.	**Limited understanding:** May struggle with complex, nuanced, or context-specific questions.
24/7 availability: Provides round-the-clock support to employees, improving accessibility.	**Dependence on data:** Requires regular updates with current company information and policies to remain accurate.
Consistency: Ensures uniform responses to common questions,	**Lack of human touch:** May lack the empathy and personal con-

reducing the chances of misinformation.	nection that human HR professionals provide.
Enhanced candidate experience: Streamlines initial interactions and scheduling during recruitment, offering timely communication.	**Potential for errors:** Incorrect responses can occur, necessitating oversight and review.
Scalability: Can handle a large volume of queries simultaneously, which is beneficial for growing companies.	**Employee resistance:** Some employees may prefer human interaction and be hesitant to use AI for HR-related queries.
Onboarding support: Guides new hires through the onboarding process, helping them settle in quickly.	**Privacy concerns:** Handling sensitive employee information requires robust data security measures to prevent breaches.
Training assistance: Recommends relevant training programs and resources, supporting employee development.	**Implementation costs:** Initial GenAI setup, model training, and ongoing maintenance can incur costs.

Harnessing GenAI in Legal

GenAI can serve as a valuable tool in the legal profession when you use it thoughtfully and with appropriate oversight. Several attorneys have found themselves in serious trouble for using AI outputs “as is.” If you’re a legal professional, you can leverage it for initial research and drafting tasks if you ensure there’s human review and verification of output.

Drafting and editing a routine legal document

For document drafting, GenAI can create first drafts of routine legal documents like basic contracts, nondisclosure agreements (NDAs), or engagement letters. For example, an attorney might ask a GenAI app such as

ChatGPT to draft a simple independent contractor agreement (see Figure 5-1) and then carefully review and modify it to ensure it meets specific jurisdictional requirements and client needs.



TIP Give GenAI a role in your prompt and then give it instructions to complete from the perspective of that role. This will render outputs more suited to how you're actually working with legal documents. This helps guide GenAI in reviewing, analyzing, responding to, or drafting documents for you. Following is an example of this type of prompt, followed by GenAI's response in Figure 5-1 (in this case, the GenAI is ChatGPT).

The prompt:

> *You are a paralegal. Draft a simple independent contractor agreement for an attorney's review.*

Independent Contractor Agreement

This Independent Contractor Agreement ("Agreement") is made and entered into on [Date] by and between [Your Company Name] ("Company") and [Contractor's Full Name] ("Contractor").

1. Services Provided:
Contractor agrees to provide the following services to the Company:
[Briefly describe services].

2. Payment:
The Company agrees to pay Contractor [amount] per [hour/project/month], payable upon [completion of services/specific date].

3. Independent Contractor Relationship:
Contractor is an independent contractor and not an employee of the Company. Contractor is responsible for all taxes, insurance, and benefits associated with their work.

4. Term and Termination:
This Agreement will begin on [start date] and continue until [end date or "completion of services"]. Either party may terminate this Agreement with [number] days' written notice.

5. Confidentiality:
Contractor agrees to keep all Company information confidential and not disclose it to any third party without prior written consent.

FIGURE 5-1: The independent contractor agreement created from a ChatGPT prompt.



WARNING I'll say it again: Never use ChatGPT or any other GenAI as the final authority on any legal document. If you're an attorney, you may want to use the chatbot to draft a document and modify it accordingly. If that's the case, note the tools revealed after you click on the toolbox icon found below the prompt bar as shown in in Figure 5-2. After the drop-down menu exposes the tools, click on Canvas (the pencil icon), which exists precisely for such editing purposes. Scroll over the section of text you want to edit and click the icon that looks like an equal sign (=) in a cartoon conversation bubble shown in Figure 5-3. The section you chose to edit will be automatically highlighted, and an editing bar will appear to the right of it, as shown in Figure 5-3. Make your edits there.

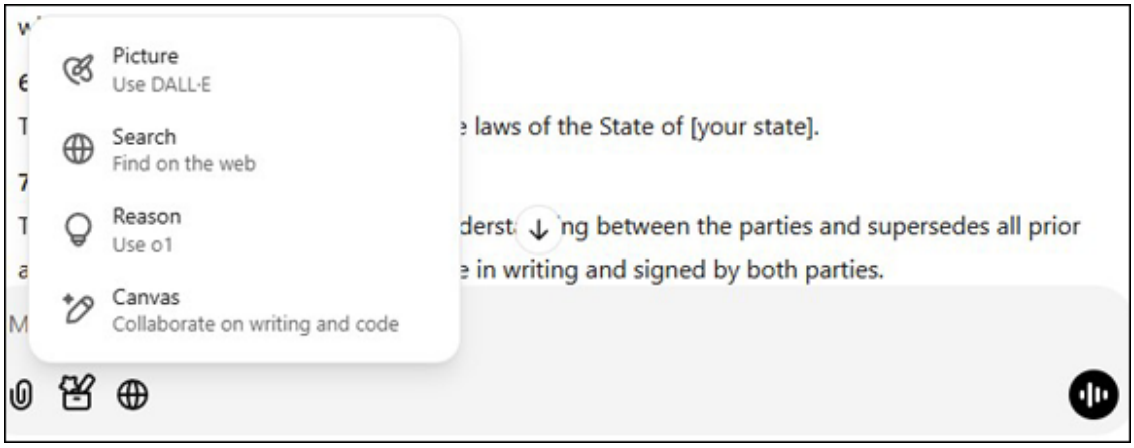


FIGURE 5-2: The toolbox under the ChatGPT prompt bar.



FIGURE 5-3: A screenshot of highlighted copy to be edited with the edit bar to the right for editing a ChatGPT response using Canvas editing tools inside of ChatGPT.

But that's not the only way I can edit this document.

Click on the icon at the bottom right of the screen (refer to Figure 5-2) to reveal another menu of tools represented by icons. Reading the icons from top to bottom:

- The first one is to Add Emojis.
- The second icon is to add Final Polish.
- The third is to adjust the Reading Level.
- The fourth is to Adjust Length.
- The final icon is another pencil icon for Suggest Edits command.

When you click on this pencil icon, you are commanding ChatGPT to review the document and Suggest Edits that you can then elect to apply or close, as seen in Figure 5-4. All of this is very helpful, but you might want to skip the Add Emojis button if you're creating an official legal document, but that button may brighten any notes you send to the paralegal for further document handling.

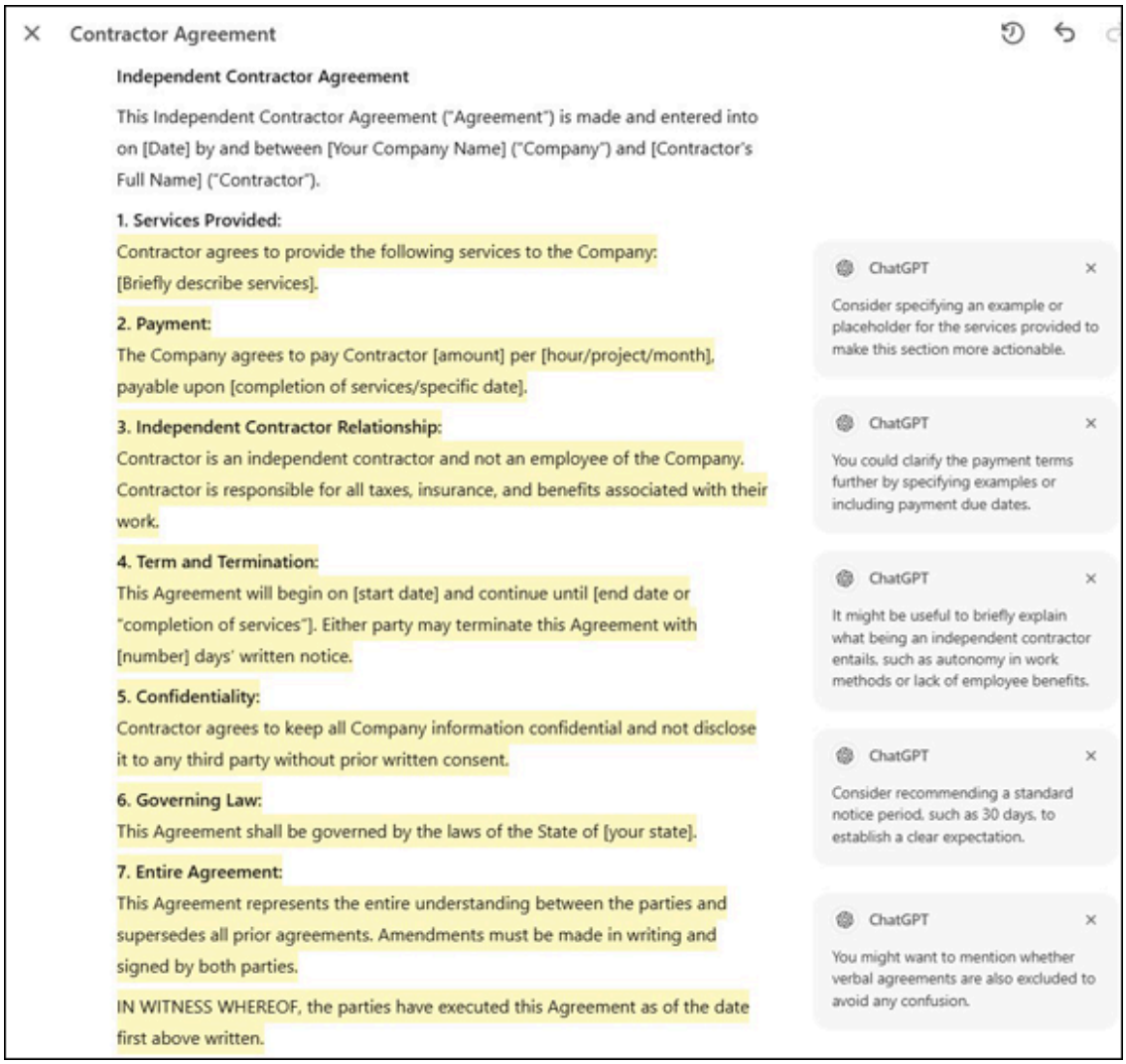


FIGURE 5-4: A screenshot of ChatGPT's suggested edits to contractor's agreement.

Researching for legal precedents and statutes

In legal research, GenAI can identify relevant cases and statutes. For example, when researching precedents for a particular type of contract dispute, GenAI might suggest relevant cases that the attorney can look up and verify. Be aware that AI tools like ChatGPT might occasionally reflect biases or make predictive errors due to the data they were trained on. This limitation is especially important for legal cases that involve sensitive issues, precedent interpretation, or potential client implications. Regularly cross-checking with trusted, up-to-date legal sources is essential.



REMEMBER Legal professionals must treat GenAI as a preliminary tool rather than a definitive source. Attorneys should independently verify all cita-

tions and legal principles through an authoritative source like Westlaw or LexisNexis.

You can expedite due diligence work using GenAI to analyze and summarize large volumes of contracts or legal documents, highlighting key terms and potential issues for human review. But, carefully verify all findings because GenAI may miss crucial details or misinterpret legal language. Although GenAI can assist in these areas, you must ensure that documents not only meet statutory requirements but also align with best practices in contract language and legal nuances that GenAI might overlook.

Drafting client responses or legal arguments

GenAI can also assist in drafting initial responses to routine client inquiries or creating explanatory materials that break down complex legal concepts into simpler terms. But don't relegate even this seemingly simple task to GenAI alone. Thoroughly review and personalize all communication before sending it to clients. Also keep in mind any policies within your firm or the jurisdiction regarding AI usage in client interactions. Some legal organizations may limit or regulate AI deployment in client-related work.

One cool and potentially valuable use of GenAI is to prompt it to help you brainstorm different legal arguments or counterarguments. If you assign GenAI different roles in your prompts, it renders multiple answers, each having a perspective that you may not have immediately considered or examined, based on the assigned role. For example, when preparing for a motion hearing as an attorney, you might use GenAI to explore potential opposing arguments and develop stronger rebuttals. Role-playing with GenAI is one of the most valuable actions you can prompt GenAI to take because it almost always delivers fresh insights for you to use.



TIP Verify all legal information and adapt any generated content to specific jurisdictional requirements and client circumstances. Using spe-

cific, detailed prompts that include relevant jurisdictional information and particular legal requirements yields more useful results.

Monitoring your GenAI use



WARNING Be extra careful with client confidentiality issues when using GenAI. You, the legal staff, and any attorney in the firm should never input privileged or confidential client information into a GenAI tool. Instead, safeguard your clients by using anonymized or hypothetical scenarios in your prompts.

Stay updated on GenAI's capabilities and limitations, and stay aware that these may fluctuate between the different underlying GPT models. Understanding what tasks GenAI can and can't reliably perform helps you and your staff maintain professional standards and ethical obligations. Compliance with any relevant guidelines or ethical standards from bar associations or firms is essential, as is understanding the limitations of AI in complex legal matters.

Look for GenAI to continue to evolve and for GenAI tools trained for the legal profession to appear regularly on the market. It's important to keep up with advancements in AI technology so you can better gauge which GenAI tool to use when and how to use GenAI most effectively overall. Regular training on AI, including data handling and prompt optimization, can help you maximize GenAI's usefulness within ethical and professional standards.

Storytelling in Journalism

GenAI can be an incredibly valuable tool for journalists. It provides assistance and support across various aspects of the work, from storytelling to data journalism to story pitches. GenAI can assist you as a journalism storyteller in several ways. Two that come to mind are generating fresh story ideas and offering unique or multiple perspectives on current events.



TIP You can streamline your workflow, speed up your writing, and expand the research and investigative aspects of your work with GenAI. The resulting prose may not be perfect, but GenAI can go a long way toward making you more productive as a journalist. However, make sure you're always using GenAI as a supplementary tool rather than a replacement for human judgment and expertise.

Exploring politics and current issues

Covering politics in any country in a nonpartisan way can be complicated when you have your own opinions on which politician you'll support. You can prompt GenAI to check a sentence or paragraph for a second opinion, or a machine evaluation as it were, on whether a statement you wrote that you know to be factual is written in a nonbiased manner. GenAI may not always get such assessments right, given that it often harbors biases, too, but it can give you other perspectives to evaluate your work by. Try giving it specific roles — several roles even — in your prompt so you can consider your interpretation of the facts from more perspectives than just your own.



WARNING Never put your entire article draft in a GenAI tool. That's especially true if you're a freelance journalist. If you do, you risk it being used to train another AI model, fine-tune the current model, or potentially being exposed in a data leak. Unless you're a GenAI tool in an enterprise version sanctioned and protected by the news organization, don't share too much information with GenAI.

That said, consider the following example of another way to use GenAI. Perhaps you're a journalist covering environmental issues. If so, you might use the prompt

> *Suggest unique angles for a story about the impact of climate change on coastal communities.*

A GenAI chatbot might suggest focusing on the economic impacts, personal stories of affected families, or innovative adaptation strategies being implemented.

Prompting AI for facts, figures, and summaries

GenAI can aid in developing a compelling narrative of the facts. For example, if you're a journalist writing a feature on a local artist, you might use the prompt

> *Help me create a compelling narrative arc for a story about a local artist's journey to success.*

A GenAI chatbot could outline key milestones and emotional beats from your notes, research, and source interviews to include in your feature story.

If you're writing hard news full of facts and numbers, GenAI can help by interpreting data analysis results and by explaining complex statistical concepts. For example, you might ask

> *Explain the significance of a p-value in data analysis.*

GenAI's explanation could be extremely helpful if you have limited statistical skills or are up against a hard deadline —which pretty much sums up the journalist experience most of the time.

GenAI can also suggest ways to visualize data. Maybe you need it to recommend specific types of charts or graphs that best represent the findings. You can use this information to create those charts and graphs in a separate visualization tool or prompt a chatbot such as ChatGPT or Copilot to make them.

If you'd prefer to add informative text to your article instead of charts or infographics, GenAI can break down intricate information into understandable narratives. For example, if you use the prompt

> *Summarize the key findings of a recent study on urban air quality.*

GenAI can distill the study's main points into an accessible summary.



TIP You can use GenAI to summarize content or to explain anything that may be confusing or esoteric in lengthy PR pitches or legal documents, company annual reports, financial reports or statements, company info on websites, and other stuff you receive or run into regularly during daily journalism work. Obviously, this won't give you all the info you need, but it will give you a strong clue as to which leads to pursue and which to take a pass on. You also might want to prioritize or organize whatever you found useful, so you know where to dig in to go deeper.

GenAI can provide general information and context to help with fact-checking, such as answering

> *What are the main health effects of prolonged exposure to air pollution?*

But make sure to cross-check the app's output with reliable sources. Hallucination and other errors will continue to be an issue with GenAI because that's the nature of these models. AI scientists can't eliminate the potential for error without rendering the model entirely useless. Accept that truth and work accordingly. Really, you have to fact-check your work — even when you don't use GenAI — because human sources can get things wrong, too.

Organizing notes and pitching a story

GenAI can assist in organizing notes and highlighting key points in your journalist notebook. For example, you might write a prompt

> *Organize my notes from an interview with a climate scientist.*

GenAI can categorize the information into themes like impacts, solutions, and personal anecdotes. You can also use GenAI to summarize lengthy interviews or documents. For instance, you can use the prompt

> *Summarize the main points from this 10-page report on renewable energy policies*

Then, GenAI can extract and condense the most pertinent information, saving you valuable time. Further, GenAI can gather background information on various topics. For example, you might ask it to

> *Provide a brief overview of the history of renewable energy development.*

From the response, you'd receive a quick summary of relevant articles, studies, or reports.

Another great use for GenAI in journalism is help with story pitches. GenAI can draft compelling pitches by suggesting a headline and a short blurb explaining the gist of the story. If you need more, the app can also provide strong opening lines and an outline of key points.

For example, you might use the prompt

> *Help me write 3 pitches for a story about the rise of urban farming from different angles for a general reader.*

GenAI can then provide three angles for that idea in three different pitches — each complete with a suggested headline and a short blurb to briefly explain the story idea. A *pitch* is essentially a sales pitch on the merits of the proposed story to an editor who may give the go-ahead to pursue the story or kill it entirely.

GenAI can also assist in customizing pitches for different editors or publications. That's especially helpful if you are a freelance journalist and want to pitch what's essentially the same idea from different angles to different kinds of publications. For example, you can prompt GenAI like this:

> *Tailor this pitch for a magazine focused on sustainable living.*

GenAI responds by adjusting the tone and focus accordingly. You can also use GenAI to refine pitches to make them more concise and impactful. For

example, from the prompt

> *Edit this pitch to make it more engaging.*

GenAI can suggest specific improvements and tighten the language.



TIP Add writer's guidelines as an attachment to your prompt to get GenAI to write a pitch more precisely tailored to that publication, website, or news outlet.

Preparing interview questions, translating, and promoting

GenAI can help you prepare for interviews by generating potential questions. For example, if you're preparing to interview a tech entrepreneur, you might use the prompt

> *Generate a list of questions for an interview with a tech startup founder.*

GenAI can provide some insightful questions and follow-up queries for you. If you want to further refine GenAI's ability to craft interview questions, try attaching documents about the company or founder to the prompt or attach an interview of a different person whom you want GenAI to use as an example of the type of information you're looking to extract in this interview.

GenAI can also help you translate text to and from various languages — although you probably should run those outputs by a human interpreter rather than just running your translated piece in a foreign outlet or a cultural community in your own country. But GenAI can be helpful in making documents and interviews more understandable in your own language so you can report on what's happening in the moment. For example, you might input this prompt:

> *Translate this interview excerpt from Spanish to English.*

GenAI can usually provide an accurate translation.

Additionally, GenAI can draft social media posts to promote your stories. Try prompting it with this:

> *Create a Tweet to promote my latest article on sustainable fashion.*

The app generates a platform-appropriate post as a response. Add a highlight or two to the prompt if you want the tweet or social post to work like a teaser or blurb for your article.

Checking your GenAI history



TIP GenAI's chat history search feature can be particularly useful for journalists. This feature allows you to quickly find earlier information or interactions you had with GenAI, making it easier to reference previous conversations without having to start from scratch.

For example, if you previously discussed a complex topic like blockchain technology with GenAI, you can use the chat history search to retrieve that information instead of asking again. You can use a prompt such as

> *Search my chat history for our previous discussion on blockchain technology.*

A prompt like this can bring up past interactions and sources, saving you a lot of time and effort.

Consulting GenAI in Healthcare

The healthcare industry is still exploring where and how GenAI might be beneficial in various aspects of patient care, medical research, and administrative tasks. Attempts so far have met with mixed results. Some GenAI deployments have been successful, and others haven't. The causes of these wins and failures are varied, as noted in the following mini-table.

Reasons for failing	*Reason for success*
Mismatched task and technology	Well-matched task and technology
Lack of AI talent and user understanding	Talented tools, users, and reviewers
Pushing the technology to perform beyond its core purpose	Establishing appropriate goals and expectations
Ill-advised dependency and data issues	Creative thinking in projects

But perhaps most defining in a solid GenAI deployment is whether the people working with it fully grasp its nature, limits, and core competencies of text generation and data discovery.

Identifying successful healthcare use cases

You might think that healthcare use cases are limited if GenAI's greatest strengths are generating text and discovering data, but you'd be mistaken. Both of those capabilities are key to an endless list of possibilities. Throw in the extra features and additional versions that keep popping up, and you end up with the means to expand the kinds of input and output modals to include things like x-rays, MRI images, and lab results like blood tests, while also sharpening the focus on the task, which may be to detect, prevent, treat, cure, or discover disease or organ system malfunctions, among others.

Now you'll look at some of the real-world use cases in which GenAI models are showing good performance:

- **Mass General Brigham:** Researchers at this institution evaluated ChatGPT's performance in clinical decision-making. ChatGPT demonstrated approximately 72 percent accuracy across all medical specialties and phases of clinical care and 77 percent accuracy in making final diagnoses. The research team's results

are published in the *Journal of Medical Internet Research.* These findings suggest the potential for ChatGPT to support clinicians in diagnostic processes.

- **Moderna:** The biotechnology company Moderna has partnered with OpenAI to integrate ChatGPT Enterprise across its operations. This collaboration aims to accelerate the development of new products by utilizing AI to optimize clinical trial doses and draft responses for regulatory submissions.
- **University of Kansas Life Span Institute:** A study that this institute conducted found that parents often trust health advice from AI tools like ChatGPT more than from healthcare professionals. This highlights the growing influence of AI in patient education and the need for careful integration into healthcare communication. Patient overreliance on information from these tools is risky.
- **Epic Systems and Microsoft:** In April 2023, Epic Systems, a leading electronic health record (EHR) company, announced a collaboration with Microsoft to integrate GPT-4 into its EHR software. This integration aims to assist healthcare providers by analyzing medical records and responding to patient inquiries, thereby enhancing efficiency, provider productivity, and patient care. It's also being seen as a way to address intense pressures on costs and margins for all healthcare institutions.



REMEMBER According to a report by the American Hospital Association, approximately half of U.S. hospitals finished 2022 with negative margins due to workforce shortages, increased labor expenses, supply disruptions, and inflationary effects. In short, increased expenses outpaced revenue increases, which then increased interest in technologies like ChatGPT and others to better that ratio.

These examples illustrate the diverse applications of GenAI in the medical field, from supporting clinical decision-making to improving patient communication and operational efficiency.

Exploring other GenAI applications in healthcare

But you can use GenAI in healthcare tasks in other ways. Here are examples to illustrate its flexibility and usefulness in specific use cases:

- **Generating multiple-choice questions (MCQs):** Educators use GenAI to create MCQs for assessments, aiding in evaluating students' understanding of medical concepts. However, the validity of AI-generated questions is under scrutiny, emphasizing the need for careful review before implementation. You can find more information on this at `https://doi.org/10.1093/postmj/qgae065`.

- **Simulating patient interactions:** GenAI serves as a virtual patient, allowing students to practice history-taking and diagnostic skills in a controlled environment. This simulation helps medical students develop communication and clinical reasoning abilities. You can find more information on these types of use cases in a paper titled "Using ChatGPT in Medical Education for Virtual Patient and Cases" by authors Meredith Ratliff, Satria Nur Sya'ban, Adonis Wazir, Sarah Haidar, and Sara Keeth.
- **Assisting in exam preparation:** Medical students employ GenAI to clarify complex topics, generate study materials, and explain challenging subjects, thereby supporting their exam readiness. You can find more information on this use case in Geeky Medics online.
- **Enhancing clinical decision-making:** GenAI aids in formulating differential diagnoses and suggesting diagnostic tests based on patient symptoms, assisting learners in considering a broader range of possibilities. However, its accuracy and reliability are still under evaluation. You can find more information on this in the paper titled "Performance of ChatGPT on USMLE: Potential for AI-Assisted Medical Education Using Large Language Models" authored by Tiffany H. Kung, Morgan Cheatham, Arielle Medenilla, Czarina Sillos, Lorie De Leon, Camille Elepaño, Maria Madriaga, Rimel Aggabao, Giezel Diaz-Candido, James Maningo, and Victor Tseng.
- **Medical education:** Educators at Harvard are incorporating ChatGPT into their curriculum to help students learn how to interact with AI in clinical settings. They use the tool to generate sample patient cases and challenge students to critically evaluate the AI-generated information. For more information about this, go to `https://magazine.hms.harvard.edu/articles/how-generative-ai-transforming-medical-education`.
- **Medical research and literature review:** Scientists at the National Institutes of Health (NIH) have experimented with using ChatGPT to generate research hypotheses and identify potential gaps in current medical knowledge, sparking new avenues for investigation. For more information on using ChatGPT in medical research, check out "Global Trends and Hotspots of ChatGPT in Medical Research: A Bibliometric and Visualized Study," found here: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11137200/`.



REMEMBER Although GenAI offers valuable tools for medical education, recognize its limitations, including potential inaccuracies and lack of clinical experience. Therefore, complement its use with traditional educational methods and oversight by qualified professionals.

Cashing In on GenAI in Finance

GenAI is used across the finance industry in several ways, but mostly to enhance customer interactions, streamline processes, and support decision-making. Banks, investment houses, insurance companies, and other finance institutions use GenAI to

- **Handle customer inquiries, troubleshoot common issues, and provide information on financial products and services in real time.** This approach reduces customer wait times, improves customer experience rankings, and allows human agents to focus on more complex tasks. For example, banks can employ GenAI to answer routine questions such as account balances and recent transactions as an instant, 24/7 service.
- **Perform investment research and analysis.** Typically, financial institutions use it to summarize complex reports and deliver insights from large volumes of financial data to different stakeholders and constituencies. Analysts and investors save time because GenAI synthesizes market information into actionable insights. For instance, investment firms use it to monitor quarterly earnings reports, highlighting performance indicators, and identifying market trends based on recent news.
- **Offer general advice on budgeting, saving, and investing** and still customizing it to fit individual goals and risk tolerance. This feature allows people without direct access to a personal financial advisor to receive guidance on managing their finances. A retirement planning app, for instance, might use GenAI to help users estimate their retirement savings or evaluate different investment options.
- **Address compliance and risk management** by using GenAI's language capabilities to interpret regulatory documents and flag potential risk. That's handy and fast and helps keep them in line with complex and evolving regulations. Its ability to quickly analyze legal texts for updates and interpret these changes in common language makes it a valuable tool for compliance teams.
  For example, GenAI might alert or explain to a bank's compliance officers a new anti-money laundering regulation or help them assess whether specific transactions meet the new reporting standards.
- **Automate routine tasks, such as content generation,** for an industry that seems to feed on an endless stream of reports and analyses. For example, a bank might use GenAI to generate regular transaction reports, identify discrepancies, or prepare presentations or summaries of financial performance over a specific period.
- **Facilitate employee training** by helping staff understand complex topics such as derivative instruments and credit risk analysis.

PROS AND CONS OF USING GENAI IN FINANCE

Pros	Cons
Efficiency: Automates routine tasks, saving time and reducing costs.	**Accuracy:** May produce errors if not properly trained or supervised.
24/7 availability: Provides round-the-clock support to customers.	**Security:** Handling sensitive financial data requires robust security measures.
Scalability: Easily scales to handle large volumes of inquiries and tasks.	**Regulation compliance:** Must ensure compliance with financial regulations and data privacy laws.
Personalization: Offers tailored advice and recommendations.	**Dependence on data quality:** Performance heavily depends on the quality of input data.
Consistency: Delivers consistent responses with human oversight.	**Lack of human touch:** May lack the empathy and nuanced understanding of human advisors.
Cost-effective: Reduces the need for large customer service teams.	**Complexity in implementation**: Integrating AI into existing systems can be complex and costly.

Using GenAI in IT Operations

GenAI is shaping up to be a valuable tool in information technology (IT) operations. This is a core department in every business, whether it's contracted as a managed service or operates on company premises. The areas it covers include things like infrastructure management, incident response, and both software and hardware troubleshooting. GenAI can help

IT streamline processes, improve efficiency, and support decision-making. IT can benefit significantly from a tool that can quickly analyze, interpret, and respond to complex data inputs.

Quite naturally then, GenAI is routinely added as a part of AIOps in many organizations. *AIOps* means "AI for IT operations" in tech speak.

IT uses GenAI

- **In cybersecurity incident reporting,** in which GenAI can assist by quickly analyzing logs and interpreting error codes. When a crash occurs — which IT would properly refer to as *system downtime* — GenAI can assist in troubleshooting the problem(s) by cross-referencing error codes with existing databases or suggesting corrective actions based on patterns learned from previous incidents. This capability saves IT time and effort and may reduce the downtime. Indeed, a case study by Forrester highlights how AI-driven tools like GenAI can improve mean time to resolution (MTTR) by assisting with automated log analysis.
- **In infrastructure management** for assisting with system monitoring and some limited automation. In a cloud environment, the app can analyze resource utilization data, such as CPU or memory usage across virtual machines, and recommend scaling actions when thresholds are met or optimization is needed.
- **To help with routine tasks** such as updating or creating content for knowledge centers and self-help repositories. It can assist help desks in similar ways. If you want specific information on how to use GenAI in IT, knowledge centers, or for help desks, consider taking one or more of my courses on LinkedIn Learning. I'm an instructor there, and my courses run the gamut on AI, IT, and help desk topics.

Overall, GenAI enhances IT operations by offering data-driven insights, providing efficient troubleshooting, and automating repetitive tasks, all of which optimize workflows and improve productivity in IT environments. However, the same warnings with other fields apply to IT. Make sure you allocate and assign individuals to provide oversight for GenAI and manage outputs carefully.

PROS AND CONS OF USING GENAI IN INFORMATION TECHNOLOGY

Aspect	Pros	Cons

Efficiency	Automates repetitive tasks (for example, retrieving data), freeing up IT teams for higher-priority tasks.	Risk of over-reliance, potentially missing nuanced issues that require human analysis.
Troubleshooting	Speeds up error diagnosis by analyzing logs and error codes and offering potential solutions based on past incidents.	Limited in handling novel issues outside its trained data scope, which may delay or mislead troubleshooting.
24/7 availability	Provides around-the-clock support, useful for global IT operations or extended hours with minimal human intervention.	Potential struggle with complex, real-time responses during critical incidents, especially without human oversight.
Resource management	Monitors system metrics (CPU, memory) and suggests scaling actions, optimizing resource usage.	Not fully equipped for decision-making in critical resource allocation without deeper contextual analysis.
Knowledge base	Serves as an extensive resource for referencing documentation and providing guidance on a wide range of technical queries.	Risk of outdated information because models need periodic updates to stay relevant with evolving technology.
Customization	Can be tailored to specific IT needs (for example, specific work-	Requires time and effort for initial setup and continuous train-

	flows, alerts) and integrated with other tools.	ing to ensure accurate and reliable responses.
User experience	Offers a user-friendly, conversational interface that's accessible to both technical and nontechnical users.	Challenges with interpreting AI-driven responses, especially for technical troubleshooting.

Examining New Businesses Based on GenAI

If you're an entrepreneur or a budding startup founder, you may want to use GenAI in a few more ways than I've outlined for your industry. To that end, consider the following guide on starting a business based on services largely generated by GenAI.

You can use GenAI to help you think of business ideas and make business plans for any type of business; your business doesn't have to sell GenAI-generated services or be built on AI from the ground up. Because I'm asked almost daily how to create a business using and selling GenAI-based services, I chose to use it here to illustrate how to use GenAI to develop a new business.

Art isn't the only thing you can generate and sell using GenAI. Imagine creating a platform that offers personalized storytelling services. For example, you can use GenAI to develop an app in which users input a few details about themselves or a loved one, and the AI generates a custom fairy tale, adventure story, or the memories of an elder family member. This can be an excellent gift in memory of a loved one or as bedtime stories for children. GenAI can develop code for you that you can use to build an app in a different platform. If you don't want to make an app, interview people or provide an online form for your customers, and then create the story by prompting GenAI to do so using that information.

Maybe you'd rather create a virtual art gallery in which other artists creating works in GenAI can display and sell their work. The gallery can also

offer a service in which customers describe their desired artwork, and their chosen artist — or you — can generate a bespoke artwork based on their description.

Similarly, you can form a content creation agency that specializes in marketing materials built by a human + GenAI team to the customer's specifications. You can even entice companies to outsource their marketing content to your agency. Businesses can provide their brand guidelines and marketing goals, and you can guide and polish GenAI blog posts, social media updates, and ad copy tailored accordingly.

Other unique opportunities are out there, too, such as in education. Consider tutoring school kids and adults alike in prompting skills or using GenAI to develop a unique study guide for students struggling with any topic. This could be particularly beneficial for homeschooling families or adult learners seeking to acquire new skills.

For the music industry, think about an AI-powered composition service. Musicians and composers can use AI to generate new melodies, harmonies, and lyrics. An entrepreneur might create a subscription-based service by which users receive regular batches of AI-generated music ideas and then refine and develop them into full compositions. It can be a valuable tool for songwriters who are experiencing creative blocks or looking for fresh inspiration.

Basic prompts to identify these and other business ideas can include questions like these:

- What are common challenges in the XYZ industry that could be alleviated or otherwise addressed with GenAI assistance?
- How can GenAI enhance personalization in my current business model?
- What creative processes in my field could benefit from GenAI-generated inspiration or content?
- In what ways can GenAI improve customer engagement and satisfaction in my business?

For a list of information that you should include in prompts for more refined or edgy business ideas, look at Figure 5-5, which also shows their importance to outcomes.

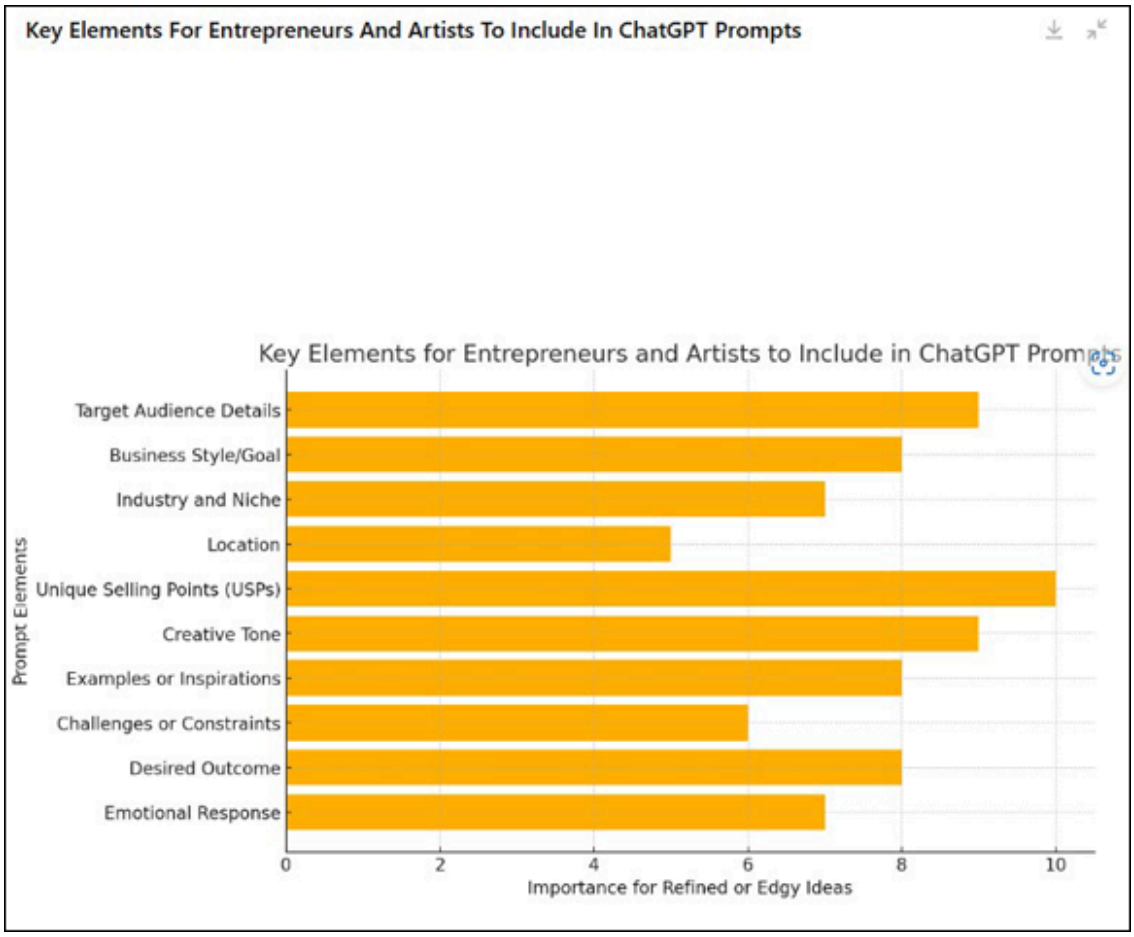


FIGURE 5-5: A list of key elements for artists and entrepreneurs to include in prompts.

Use your imagination and explore all you want. Odds are that GenAI outputs will spark some great ideas.